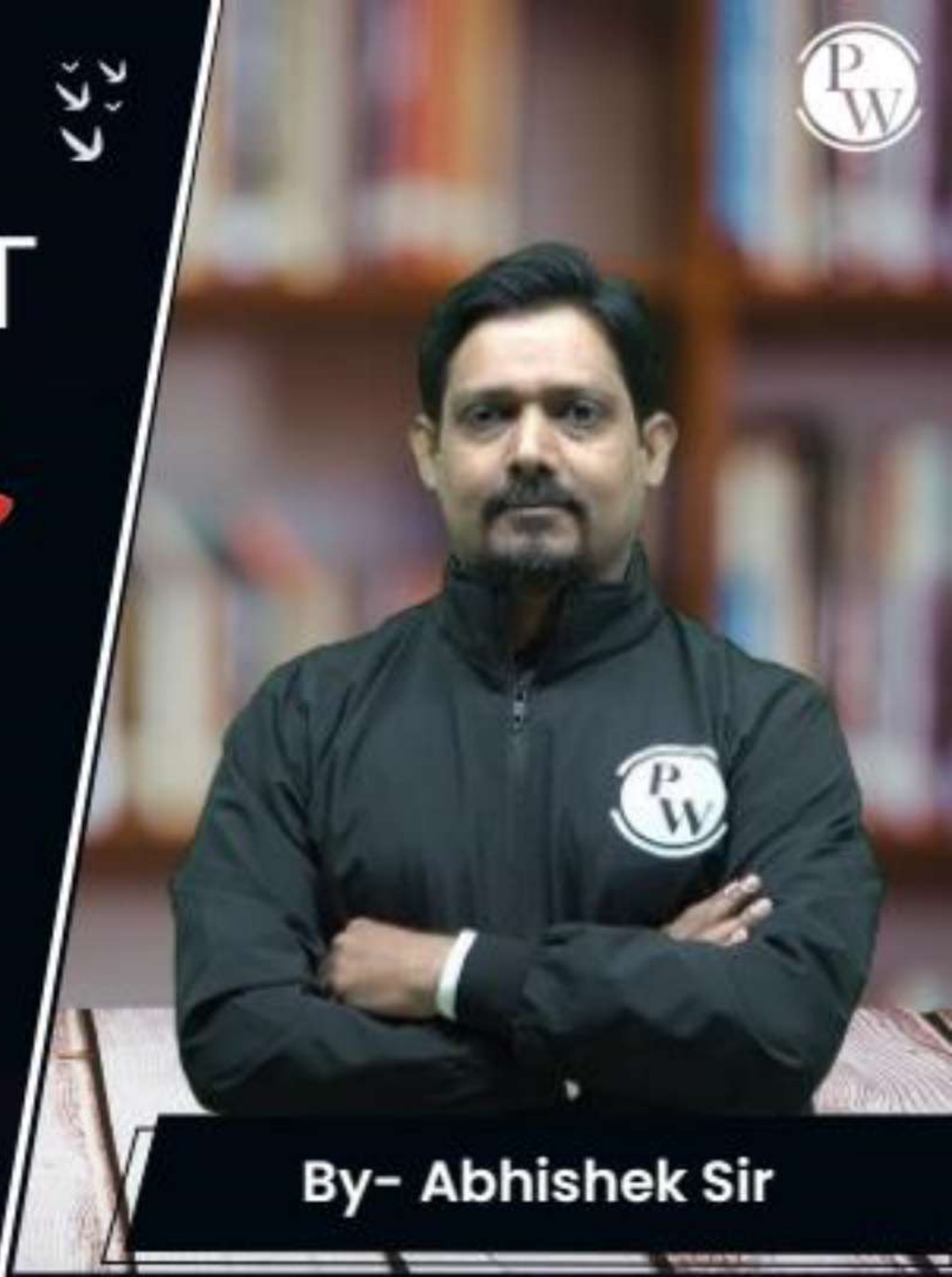
Computer Science & IT



C Programming

Control Flow Statements

Lecture No. 02



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Recap of Previous Lecture



- **Topic** Printf function
- **Topic** Ternary function
- **Topic** Control Statements
- **Topic**
- **Topic**

Slide

Topics to be Covered



- Topic: If else
- Topic: If else if
- Topic: Switch ✓ Menu.
- Topic
- Topic

Slide



GATE 2017

1 mark (NAT) ++ operator modify

Consider the following C program.

```
# include <stdio.h>
int main () {
      int m = 10;
      int n, n1;
      n = ++m;
      n1 = m++;
      n--;
      --n1;
      n - = n1;
      printf ("%d", n) ;
      return 0;
}
```

m | 10 | 11 | 12

n | 11 | 10

n_1 | 11 | 10

post increment

$n = n - n_1 \Rightarrow n = 10 - 10 = 0$

0

The output of the program is 0 ✓.

Slide

$n -= n_1 \Rightarrow$ Short form $n = n - n_1;$

$n += n_1 \Rightarrow n = n + n_1$ $n += 3$

$n *= n_1 \Rightarrow n = n * n_1$

$n /= n_1 \Rightarrow n = n / n_1$

$n = n + 3$



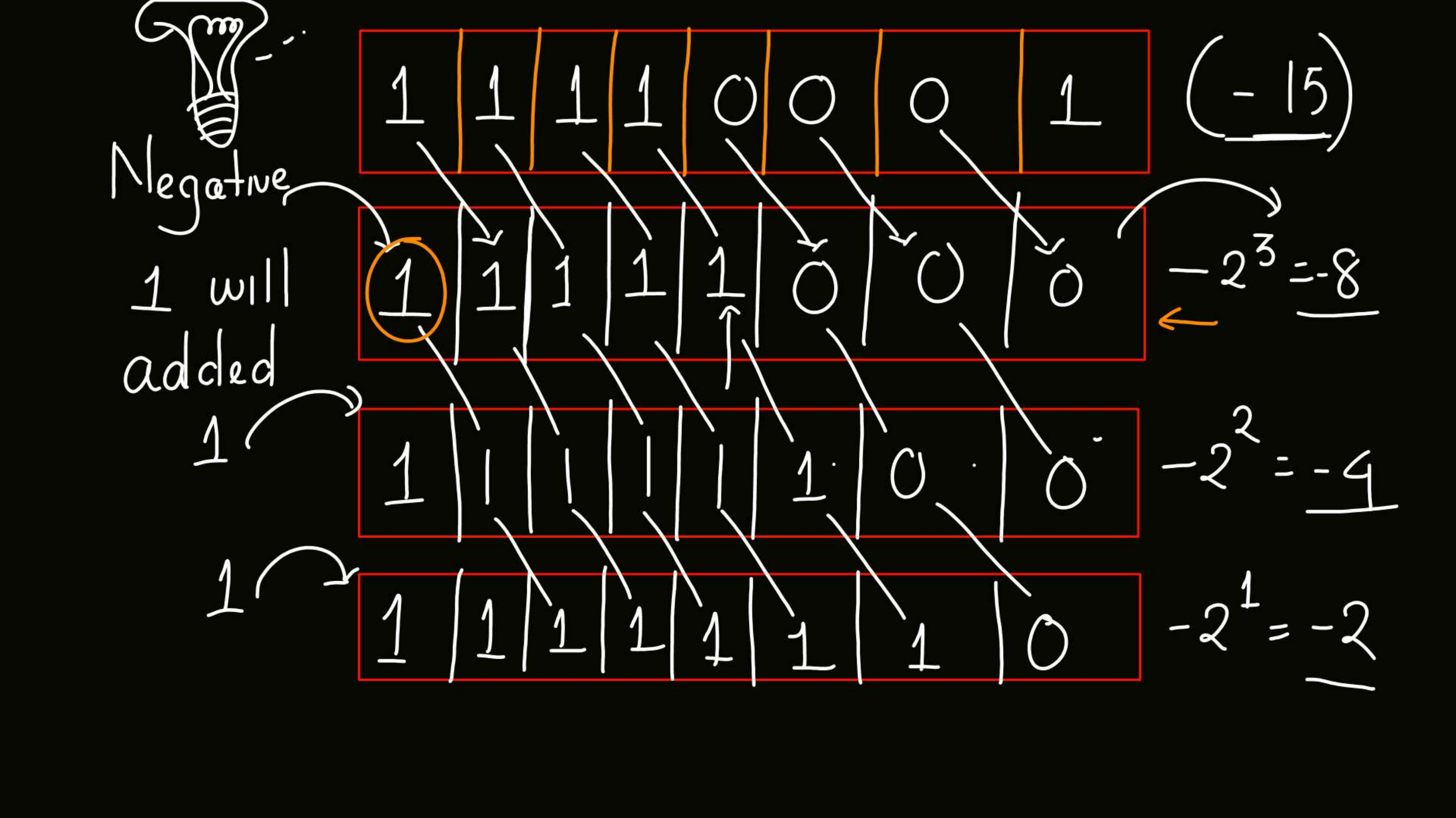
Bit-wise Operator

```
#include<stdio.h>
   int main(){
            char a = -15;
            int k;
            k =a>>3;
            printf("%d", k);
            return 0;
   }
```

-2

(A) -1
(B) -2
(C) 4
(D) 6

Slide


1 1 1 1 0 0 0 1
(-15)
Negative
1 will added
1 1 1 1 1 0 0 0
$-2^3 = -8$
1
1 1 1 1 1 1 0 0
$-2^2 = -4$
1
1 1 1 1 1 1 1 0
$-2^1 = -2$

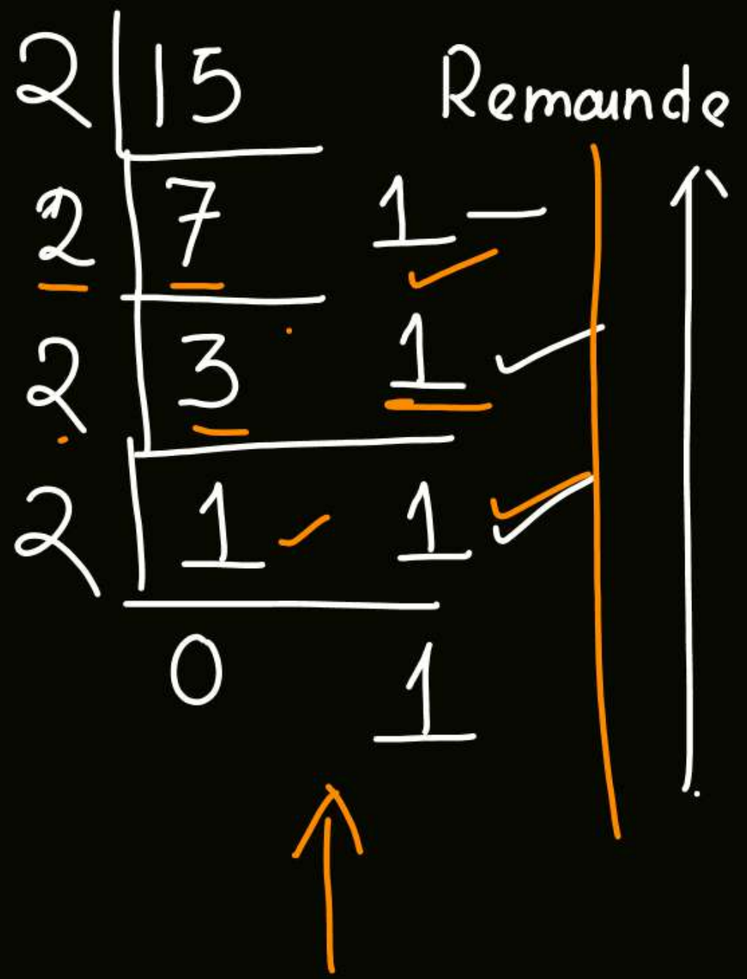
Left shift zeros will be added.

00001111 (+15)

11110000 (1's complement)

1

11110001 (-15)



1 shift $\overline{\uparrow}\left\lceil \frac{15}{2} \right\rceil - \lceil 7.5 \rceil = -8$ Not sure

3-shift $\overline{\uparrow}\left\lceil \frac{15}{2^3} \right\rceil = -\lceil 1.8 \rceil = -2$

Negative

toy karo

Not sure

$-\left\lceil \frac{10}{2} \right\rceil = -5$

$-\left\lceil \frac{5}{2} \right\rceil = -3$ $-\left\lceil \frac{3}{2} \right\rceil = -2$



Question

```
#include <stdio.h>
int main (){

    int i=0;
    if(i++){
        printf("Namaskar");
    }
    printf("\t Dunia");
    return 0;
}
```

post increment

if(0)

in this expression i value will be 0

(i++) / if(0)

(A) NamaskarDunia

(B) Namaskar Dunia

(C) Namaskar

(D) Dunia

Slide



Question

[B]

if(0)

```
#include <stdio.h>
int main (){

    int a=0;
    if(a++);
    {
        a = a+10;
        a = a-5 ;

    }
    printf("%d", a);
    return 0;

}
```

Will execute

Semicolon

if statement terminated

a = 1

post increment

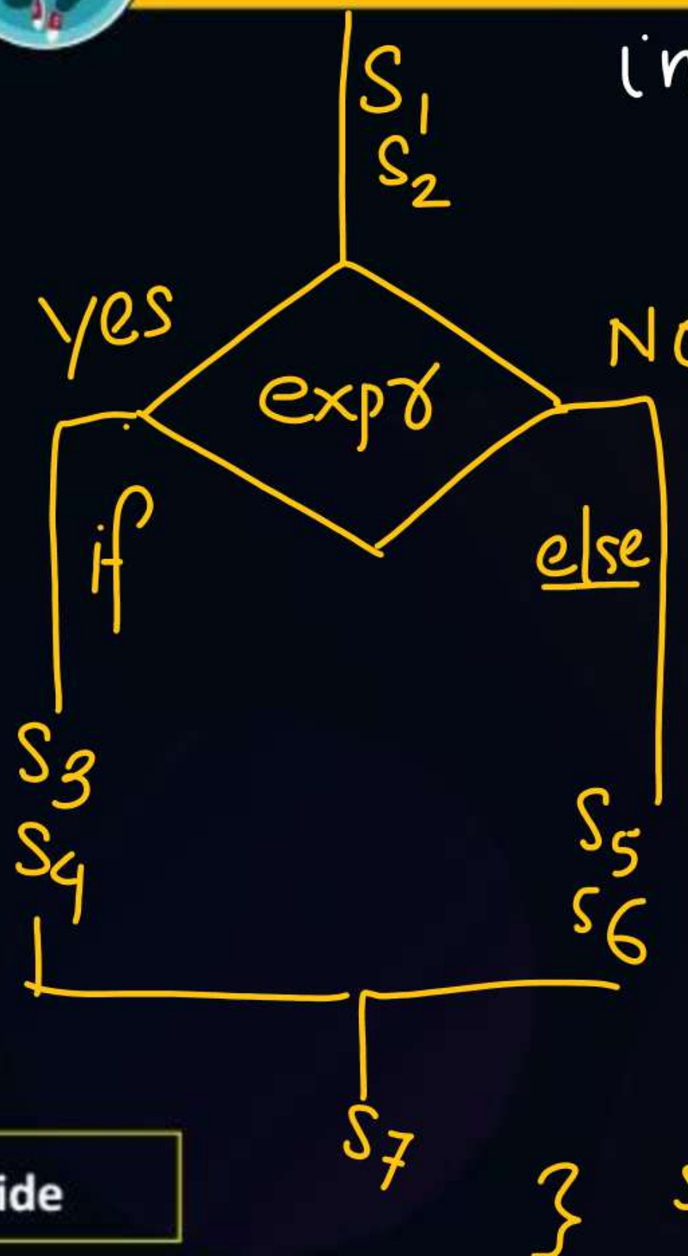
a = 1 + 10 = 11

a = a - 5 11 - 5 = 6

(A) 5 (B) 6 (C) 15 (D) 20

Slide

If Else Statement




```
int main() {
    S1
    S2
    if (expr) {
        S3
        S4
    } else {
        S5
        S6
    }
    S7;
}
```

expression. Nonzero → then S_3 & S_4

Zero

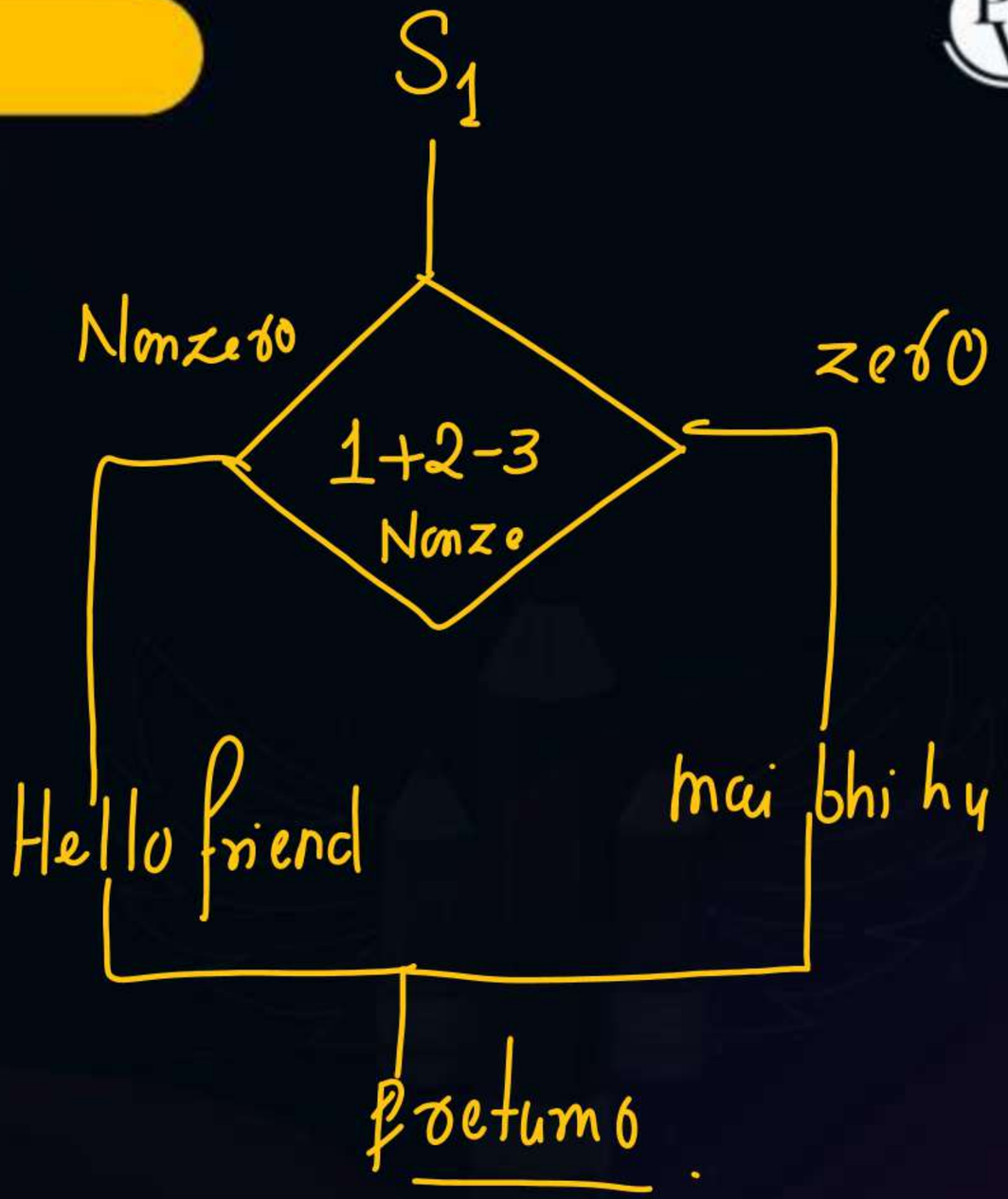
Slide



If else Statement

```
int main(){
      s1;
      if(condition/expression){
            s2;
            s3;
      }
      else {
            s4;
            s5;

      }
      s6;
}
```





Question

[B]

```
#include <stdio.h>
     int main (){

          int i=1;
          if(i+2-3){
               printf("Hello friends");
          }
          else
               printf("mai bhi hu");
          return 0;
}
```

(a) Hello friends
(b) mai bhi hu ✓
(c) Hello frinds mai bhi hu
(d) No Output

i+2-3 → expression

1+2-3

3-3

0



Question

```
#include <stdio.h>
     int main (){

          int i=0;
          if(i++){
               printf("Hello friends");
          }
          else
               printf("mai bhi hu");
          return 0;
}
```

[B]

(a) Hello friends
(b) mai bhi hu
(c) Hello frinds mai bhi hu
(d) No Output

[B]

post increment



Question

```
#include <stdio.h>
     int main (){
          int i=1;
          if(i++){
               printf("Hello friends");
          }
          else;
               printf("\tmai bhi hu");
          return 0;
}
```

(a) Hello friends
(b) mai bhi hu
(c) Hello frinds mai bhi hu
(d) No Output

Number is Even or Odd



if

else

```
int a;
printf("ENTER The No.");
scanf("%od", &a);
if (a%2==0){
        printf("the given No %d is Even", a)
}else{
        printf("the given No %d is odd", a);
}
return 0;
```

Slide



Number is Even or Odd

```
int main() {
    int num;
    printf("Enter an integer: ");
    scanf("%d", &num);

    // true if num is perfectly divisible by 2
    if(num % 2 == 0)
        printf("%d is even.", num);
    else
        printf("%d is odd.", num);

    return 0;
}
```



Even Odd, ternary Operator

Clear?

```
int a;
printf("Enter the No");
scanf("%d", &a);
(a%2)==0 ? printf("No is Even") : printf
("No. is odd");
```

Slide



Even Odd, ternary Operator

```
#include <stdio.h>
      int main() {
       int num;
       printf("Enter an integer: ");
       scanf("%d", &num);
      (num%2==0)?printf("%d is even.", num):printf("%d is odd.", num);
      return 0;
      }
```



Home Work

```
Decide the grade based on the marks
    If marks <50 then Grade is F
    if marks >=50 <60 then Grade is D
    if marks >=60 <70 then Grade is C
    if marks >=70 <80 then Grade is B
    if marks >=80 <90 then Grade is A
    if marks >=90 then Grade is A+
```

if else if

Grading System

1 user input marks

if (mark < 50) ✓

grade is F

else if (marks >= 50 && mark < 60)

Grade D

else

Slide



Switch statements

Control statement

flow of if else - Linear flow

Switch statement allows a a value to be tested against List of values.

Each List value is called a case.

Slide



Application

Menu option create

switch (expr) {

Case 1 : _____

Case 2 : _____

Case 3 : _____

default : _____

}

block

Constant Expression

No variable

default case will be executed
If No case evaluates to true.



Switch Syntax

```
switch(expression){
   case constant-expression  :
      statement(s);
      break; //optional
   case constant-expression  :
      statement(s);
      break; //optional

   // you can have any number of case statements.
   default : //Optional
      statement(s);
}
```

Const
constant expression

operation

break;



If No break statement

Switch
Case
default
break;

break Statement

break : Is Not mandatory

break keyword transfers the control outside of switch block.

Slide



Question

```
#include <stdio.h>
    int main(){
     int a = 5;
      switch(a){
              case 4: printf("%d",5);
              case 5: printf("%d",5);
              case 6: printf("%d",6);
      }
    }
```

a = 0

a = 5

Neither default Nor break

5 will be tested for equality against

if break present then it will print 5

56

Slide

[default and break are optional]

if break is Not present then after that case every case evaluates to true.



Question

```
#include <stdio.h>
      int main(){
       int a = 5;
        switch(a-1){
                  case a: printf("%d",5);
                  case 5: printf("%d",5);
                  case 6: printf("%d",6);
        }
      }
```



Question

```
#include <stdio.h>
int main(){


    switch(13/4){
        case 4: printf("%d",4); break ;
        case 3: printf("%d",2); break ;
        case 5: printf("%d",5); break ;
}
    }
```



GATE 2012

What will be the output of the following C program segment?

```
Char inChar = 'A' ;
switch (inChar ) {
case 'A' : printf ("Choice A\ n");
      case 'B' :
      case 'C' : printf("Choice B");
      case 'D' :
      case 'E' :
      default  : printf ("No Choice") ; }
```

(A) No choice
(B) Choice A
(C) Choice A
Choice B No choice
(D) Program gives no output as it is erroneous

Slide



2 mins Summary

- Topic
- Topic
- Topic
- Topic
- Topic

Slide

THANK - YOU